Exponential STFT concentration does not force a Poincaré inequality

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Source and conjecture

The source is Martin Rathmair, *Stable STFT phase retrieval and Poincaré inequalities* [1]. It is available as arXiv:2407.00398. Its short-time Fourier transform convention is

$$\mathcal{V}_g f(x,\xi) = \int_{\mathbb{R}} f(t)\overline{g(t-x)}e^{-2\pi i\xi t}\,dt.$$

Conjecture 1.18 on page 7 fixes

$$g(t) = e^{t-e^t}, \qquad \gamma(x,\xi) = e^{-a|x|-b|\xi|} \quad (a,b>0),$$

and asks whether

$$\exists\delta>0: \quad |\mathcal{V}_g f(z)|e^{\delta|z|} \in L^\infty(\mathbb{R}^2) \quad \Longrightarrow \quad C_P\big((|\mathcal{V}_g f|^2 * \gamma)^2\big) < \infty.$$

Here $C_P(w)$ is the least constant in

$$\inf_{c\in\mathbb{C}} \int_{\mathbb{R}^2} |u-c|^2 w \le C_P(w)\int_{\mathbb{R}^2} |\nabla u|^2 w, \qquad u \in C_b^\infty(\mathbb{R}^2).$$

1.4.2. *SLPR beyond STFT.* So far we only know how to establish Γ-SLPR when $V = \mathcal{V}_g(L^2(\mathbb{R}))$ arises as the image of the STFT (and for very particular windows g only). For that case we have rather explicit knowledge what the relevant extension operator looks like (see Section 2.3). From this perspective it is natural to ask:

> Can the general result, Theorem 1.9 be applied beyond the STFT setting? Can the property of SLPR be established for any other natural function classes V?

1.4.3. *Concentration implies finite Poincaré constant?* Any weight w which supports a Poincaré inequality has exponential concentration, cf. [2, Theorem 2]. The converse is of course not true: Just consider a weight which has compact but disconnected support. In our situation such weights are however not permitted as $w = (|F|^2 * \gamma)^2$ is strictly positive. One may therefore ask:

> Is exponential concentration of $F = \mathcal{V}_g f$ sufficient for $w = (|F|^2 * \gamma)^2$ to support a Poincaré inequality?

More formally, this can be phrased as follows:

Conjecture 1.18. *Let $g(t) = \exp(t - e^t)$, and let $\gamma(x, \xi) = \exp(-a|x| - b|\xi|)$, $a, b > 0$. Suppose that $f \in L^2(\mathbb{R})$ has exponential time-frequency concentration, i.e.,*

$$\exists \delta > 0 : \quad |\mathcal{V}_g f(z)| \cdot e^{\delta |z|} \in L^\infty(\mathbb{R}^2)$$

and let $w = (|\mathcal{V}_g f|^2 * \gamma)^2$. Then $C_P(w) < \infty$.

Proof intuition

Strict positivity removes literally disconnected supports, but it does not remove arbitrarily thin bottlenecks. The ambiguity function $A = \mathcal{V}_g g$ is exponentially localized. We place copies of g at the lacunary positions $R_n = L2^n$, with coefficients $e^{-\kappa R_n}$. Their exponential decay is enough to preserve the hypothesis of the conjecture. On the other hand, the smoothed spectrogram has a fixed-size island near every R_n, while its value halfway from R_{n-1} to R_n has an additional exponentially small factor. Step functions changing value at those midpoints have non-negligible variance on the n-th island and vanishingly smaller Dirichlet energy in the bottleneck. Hence no uniform Poincaré constant exists.

Counterexample

Write $T_R h(t) = h(t - R)$ and $A = \mathcal{V}_g g$. The source computes

$$|A(x, \xi)|^2 = \frac{1}{16} \operatorname{sech}^4(x/2) |\Gamma(2 - 2\pi i \xi)|^2. \tag{1}$$

Put $H(\xi) = |\Gamma(2 - 2\pi i \xi)|$. In particular,

$$|A(x, \xi)| \le C e^{-|x|} H(\xi), \qquad \sup_{\xi \in \mathbb{R}} e^{d|\xi|} H(\xi) < \infty \quad (0 < d < \pi^2). \tag{2}$$

The second assertion also follows immediately from the standard Gamma-function identity used in the source.

Theorem 1. *For every $a, b > 0$, there is a nonzero $f \in L^2(\mathbb{R})$ such that*

$$|\mathcal{V}_g f(z)| e^{\delta|z|} \in L^\infty(\mathbb{R}^2)$$

for some $\delta > 0$, but, with $\gamma(x,\xi) = e^{-a|x|-b|\xi|}$ and $w = (|\mathcal{V}_g f|^2 * \gamma)^2$, one has $C_P(w) = \infty$. Consequently Conjecture 1.18 is false for every parameter pair $a, b > 0$.

Proof. Fix $r \in (0,2)$, $s \in (0, \min\{a, r\})$, and

$$0 < \kappa < s/4. \tag{3}$$

For a constant $L > 0$, to be chosen sufficiently large, set

$$R_n = L2^n, \qquad c_n = e^{-\kappa R_n}, \qquad f = \sum_{n=1}^\infty c_n T_{R_n} g.$$

Since $\sum_n c_n < \infty$, this series converges in $L^2(\mathbb{R})$. STFT covariance gives

$$F(x,\xi) := \mathcal{V}_g f(x,\xi) = \sum_{n=1}^\infty c_n e^{-2\pi i \xi R_n} A(x - R_n, \xi). \tag{4}$$

The concentration hypothesis holds. By (2),

$$|F(x,\xi)| \le C H(\xi) S(x), \qquad S(x) := \sum_n c_n e^{-|x-R_n|}.$$

If $0 < \delta < \min\{\kappa, 1, \pi^2\}$, then

$$e^{\delta|x|} S(x) \le \sum_n c_n e^{\delta R_n} e^{-(1-\delta)|x-R_n|} \le \sum_n e^{-(\kappa-\delta)R_n} < \infty.$$

Together with (2) and $e^{\delta|(x,\xi)|} \le e^{\delta|x|} e^{\delta|\xi|}$, this proves

$$|F(z)| e^{\delta|z|} \in L^\infty(\mathbb{R}^2).$$

An upper bound in the bottlenecks. The centers R_n are uniformly separated. Weighted Cauchy–Schwarz gives

$$\begin{aligned} S(x)^2 &\le \left(\sum_n c_n^2 e^{-r|x-R_n|}\right)\left(\sum_n e^{-(2-r)|x-R_n|}\right) \\ &\le C_r \sum_n c_n^2 e^{-r|x-R_n|}. \end{aligned} \tag{5}$$

Let

$$q = |F|^2 * \gamma, \qquad w = q^2.$$

Convolving (5) separately in x and ξ, and using

$$e^{-r|\cdot|} * e^{-a|\cdot|} \le C_{r,a,s} e^{-s|\cdot|},$$

we obtain

$$q(x,\xi) \le C B(\xi) P(x), \qquad B = H^2 * e^{-b|\cdot|} \in L^2(\mathbb{R}), \quad P(x) = \sum_n c_n^2 e^{-s|x-R_n|}. \tag{6}$$

The L^2-claim for B follows from the exponential decay of H and Young's inequality.

For $n \geq 2$, let

$$M_n = \frac{R_{n-1} + R_n}{2} = \frac{3}{2} R_{n-1}, \qquad I_n = [M_n - 1, M_n + 1].$$

An elementary comparison of the terms in P shows that, uniformly for $x \in I_n$,

$$P(x) \leq C c_n^2 e^{-\eta R_{n-1}}, \qquad \eta := s/2 - 2\kappa > 0. \tag{7}$$

Indeed, among $j < n$ the largest term is $j = n - 1$, because $s - 2\kappa > 0$, and

$$\frac{c_{n-1}^2 e^{-s(M_n - R_{n-1} - 1)}}{c_n^2} = e^{-(s/2 - 2\kappa) R_{n-1} + s}.$$

Among $j \geq n$ the largest term is $j = n$, and its ratio to c_n^2 is at most $e^{-sR_{n-1}/2+s}$. The remaining lacunary tails are bounded geometric sums. From (6)–(7),

$$\int_{I_n \times \mathbb{R}} w(x, \xi)\, dx\, d\xi \leq C c_n^4 e^{-2\eta R_{n-1}}. \tag{8}$$

A lower bound on every island. Since $A(0,0) \neq 0$, choose a small rectangle $Q = [-\varepsilon, \varepsilon]^2$ and $m > 0$ such that $|A| \geq m$ on Q. Increasing L, if necessary, makes the principal term in (4) dominate all other translates, uniformly in n:

$$|F(x, \xi)| \geq \frac{m}{2} c_n \quad \text{when } (x - R_n, \xi) \in Q.$$

To see this, divide the upper bound for every other term by c_n. For $j < n$ it is, up to a fixed factor, $e^{-(1-\kappa)(R_n - R_j)}$, while for $j > n$ it is $e^{-(1+\kappa)(R_j - R_n)}$. Both lacunary sums tend uniformly to zero as $L \to \infty$.

Let Q' be a smaller concentric rectangle. For $E_n = (R_n, 0) + Q'$, integrate the convolution defining q over $(R_n, 0) + Q$. The kernel γ has a fixed positive lower bound on the relevant bounded difference set, and hence

$$q \geq d_0 c_n^2 \ \text{ on } E_n, \qquad \int_{E_n} w \geq d_1 c_n^4 \tag{9}$$

with constants $d_0, d_1 > 0$ independent of n.

The Poincaré quotients diverge. The function q lies in $L^1 \cap L^\infty$: Moyal's identity gives $|F|^2 \in L^1$, while $\gamma \in L^1 \cap L^\infty$. Thus $w = q^2 \in L^1$. It is positive and nonzero; write $m_w = \int_{\mathbb{R}^2} w \in (0, \infty)$.

Choose $\theta \in C^\infty(\mathbb{R}, [0,1])$ with $\theta(t) = 0$ for $t \leq -1$, $\theta(t) = 1$ for $t \geq 1$, and bounded derivative, and put $u_n(x, \xi) = \theta(x - M_n)$. By (8),

$$\int_{\mathbb{R}^2} |\nabla u_n|^2 w \leq C c_n^4 e^{-2\eta R_{n-1}}. \tag{10}$$

For every finite measure $d\mu = w\, dz$,

$$\inf_c \int |u - c|^2\, d\mu = \frac{1}{2m_w} \iint |u(z) - u(z')|^2\, d\mu(z) d\mu(z').$$

The mass of $\{x \leq M_n - 1\}$ tends to m_w, whereas $u_n = 1$ on $E_n \subset \{x \geq M_n + 1\}$ for large n. Restricting the double integral to these two sets and using (9) yields

$$\inf_c \int_{\mathbb{R}^2} |u_n - c|^2 w \geq \frac{d_1}{2} c_n^4 \tag{11}$$

for all sufficiently large n. Dividing (11) by (10) gives a quotient bounded below by $Ce^{2\eta R_{n-1}} \to \infty$. Therefore $C_P(w) = \infty$, completing the counterexample. $\square$

Verification notes

The construction is analytic and uses no numerical or unproved dependency. The parameter inequalities are simultaneously satisfiable for every $a, b > 0$: choose $r < 2$, then $s < \min\{a, r\}$, then $\kappa < s/4$, and finally L large. The three review-critical estimates are (5), the midpoint comparison (7), and the uniform main-translate dominance preceding (9).

Novelty check and limitations

On 2026-08-11, searches for the exact paper title, the phrases `‘‘Concentration implies finite Poincare constant’’` and `‘‘exponential time-frequency concentration’’`, and `Conjecture 1.18` together with STFT/Poincaré terminology found the source arXiv and published versions but no later claimed resolution. The local arXiv source corpus and the run's four lightweight indexes likewise had no matching result. This is a bounded novelty check, not a claim that every non-arXiv source has been exhausted.

The theorem disproves the conjecture for all its parameter values. It does not characterize additional structural assumptions on f (for example, unimodality or log-concavity of the spectrogram) that would restore a finite Poincaré constant.

Human review recommendation

Review as a likely-valid full counterexample. The most useful independent check is to verify the exponents in (7): the previous island contributes $c_{n-1}^2 e^{-s(R_n - R_{n-1})/2}$, while the target island has mass scale c_n^2; the choice $\kappa < s/4$ makes their ratio decay.

References

[1] Martin Rathmair, *Stable STFT phase retrieval and Poincaré inequalities*, arXiv:2407.00398; International Mathematics Research Notices 2024, no. 22, 14094–14114.